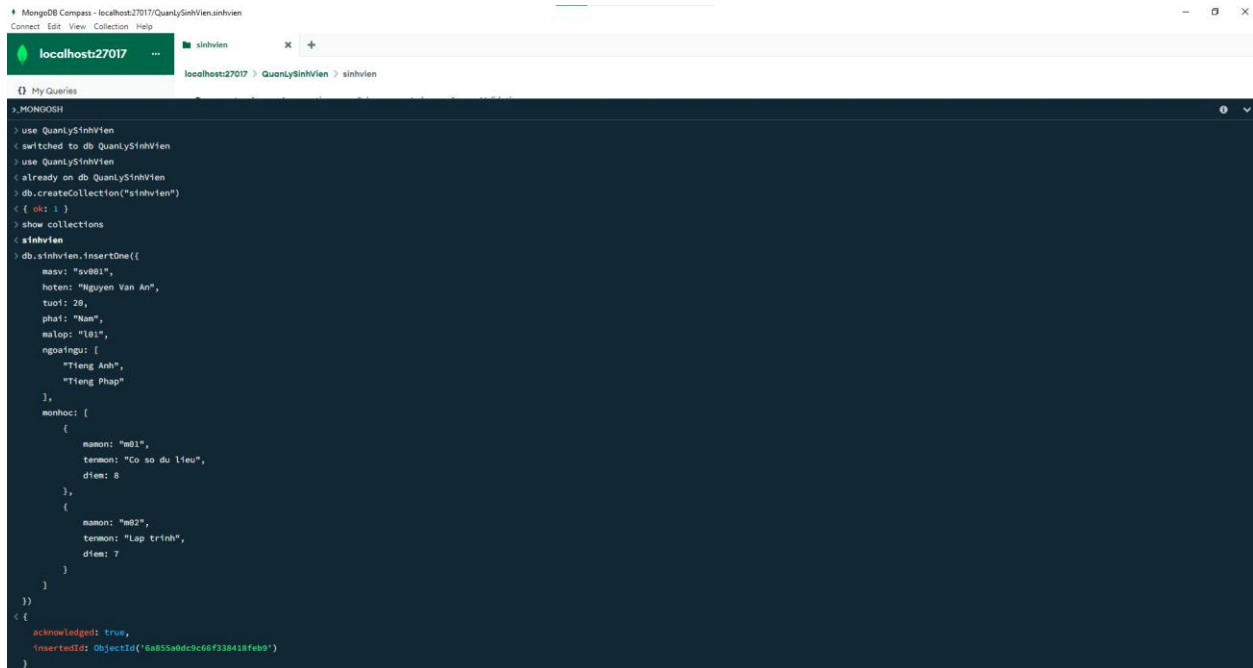
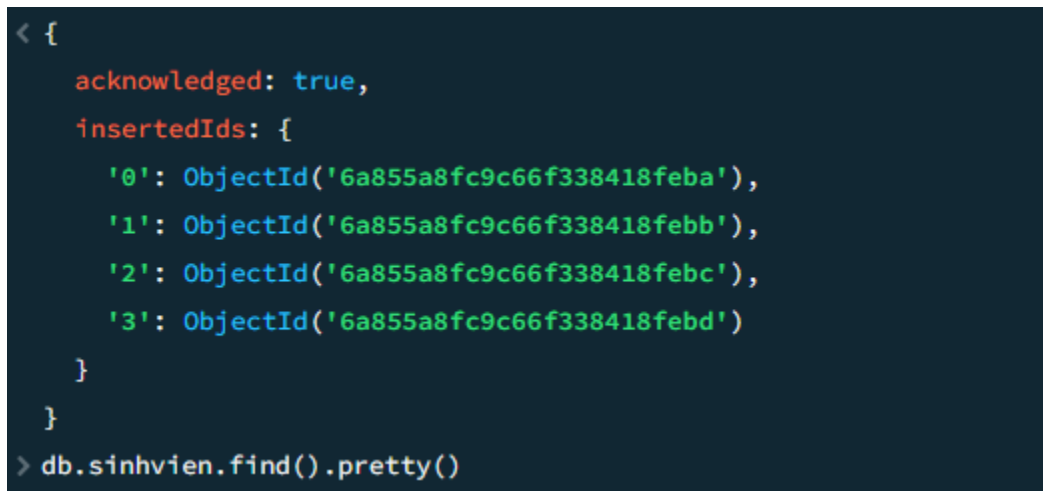


a)



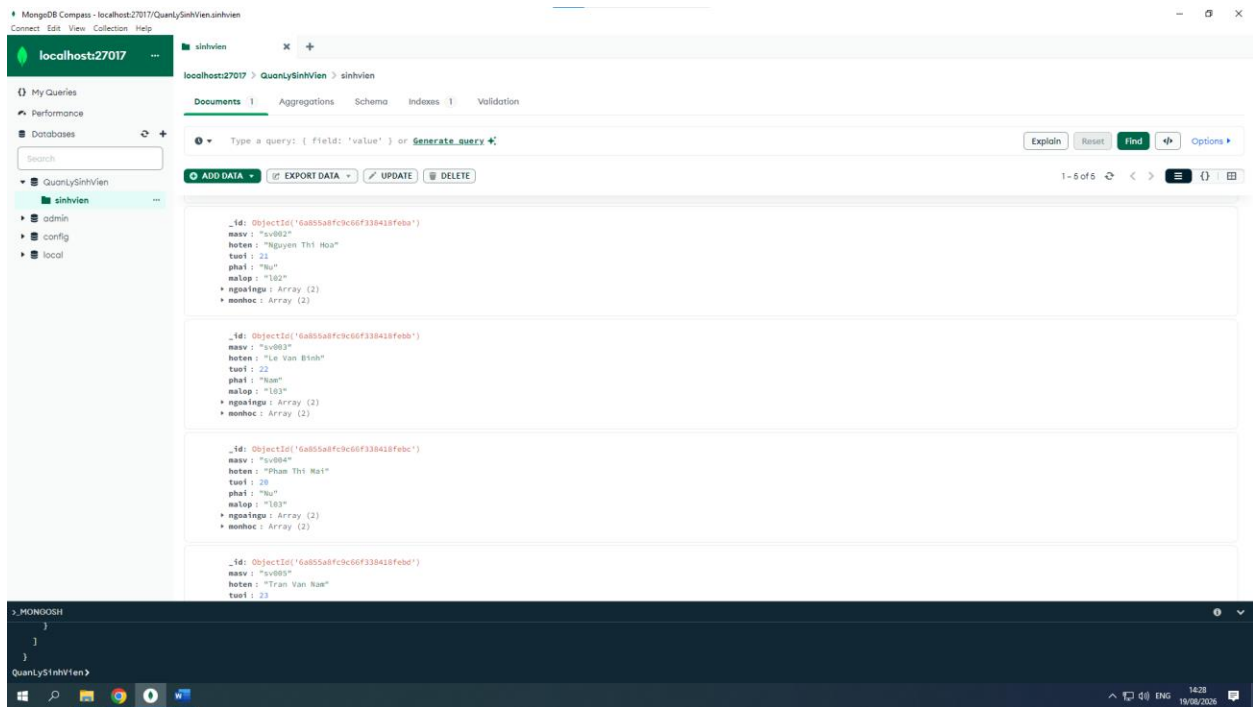
The screenshot shows the MongoDB Compass application window. The title bar reads "MongoDB Compass - localhost:27017/QuanLySinhVien.sinhvien". Below the title bar is a menu bar with "Connect", "Edit", "View", "Collection", and "Help". The main interface has a sidebar on the left with a green button labeled "localhost:27017" and a tab labeled "sinhvien". The main area displays the MongoDB shell output for the "sinhvien" collection. The output shows the following commands and results:

```
> use QuanLySinhVien
switched to db QuanLySinhVien
> use QuanLySinhVien
already on db QuanLySinhVien
> db.createCollection("sinhvien")
{ ok: 1 }
> show collections
sinhvien
> db.sinhvien.insertOne({
  masv: "sv001",
  hoten: "Nguyen Van An",
  tuoi: 20,
  phai: "Nam",
  malop: "L01",
  ngoaingu: [
    "Tieng Anh",
    "Tieng Phap"
  ],
  monhoc: [
    {
      monon: "m01",
      tenmon: "Co so du lieu",
      diem: 8
    },
    {
      monon: "m02",
      tenmon: "Lap trinh",
      diem: 7
    }
  ]
})
{
  acknowledged: true,
  insertedId: ObjectId('6a855a8dc66f338418feb9')
}
```



The screenshot shows the MongoDB shell output for the command `db.sinhvien.find().pretty()`. The output is a JSON object with the following structure:

```
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a855a8fc9c66f338418feba'),
    '1': ObjectId('6a855a8fc9c66f338418febb'),
    '2': ObjectId('6a855a8fc9c66f338418febc'),
    '3': ObjectId('6a855a8fc9c66f338418febd')
  }
}
```



b)

```
> db.sinhvien.deleteOne({
  masv: "sv005"
})
< {
  acknowledged: true,
  deletedCount: 1
}
```

MongoDB Compass - localhost:27017/QuanLySinhVien.sinhvien

localhost:27017

sinhvien

Documents | Aggregations | Schema | Indexes | Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA | EXPORT DATA | UPDATE | DELETE

1 - 4 of 4

```
{
  "_id": "sv001",
  "masv": "sv001",
  "hoten": "Le Van Binh",
  "tuoi": 22,
  "phai": "Nam",
  "malop": "182",
  "ngoaingua": Array (2),
  "monhoc": Array (2)
}
```

```
{
  "_id": "sv002",
  "masv": "sv002",
  "hoten": "Pham Thi Huu",
  "tuoi": 20,
  "phai": "Nu",
  "malop": "183",
  "ngoaingua": Array (2),
  "monhoc": Array (2)
}
```

```
> db.sinhvien.deleteOne({
  masv: "sv001"
})
{
  acknowledged: true,
  deletedCount: 1
}
```

QuanLySinhVien

c)

MongoDB Compass - localhost:27017/QuanLySinhVien.sinhvien

localhost:27017

sinhvien

Documents | Aggregations | Schema | Indexes | Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA | EXPORT DATA | UPDATE | DELETE

1 - 2 of 2

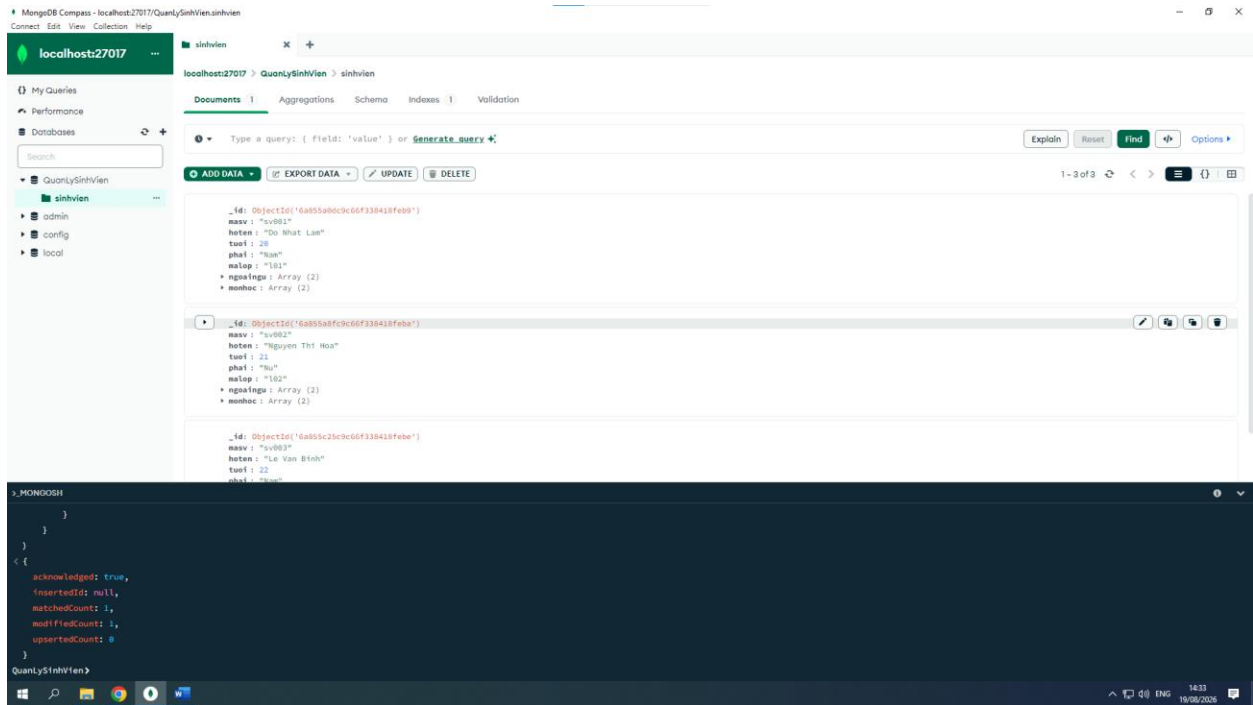
```
{
  "_id": "sv001",
  "masv": "sv001",
  "hoten": "Nguyen Van An",
  "tuoi": 20,
  "phai": "Nam",
  "malop": "181",
  "ngoaingua": Array (2),
  "monhoc": Array (2)
}
```

```
{
  "_id": "sv002",
  "masv": "sv002",
  "hoten": "Nguyen Thi Hoa",
  "tuoi": 23,
  "phai": "Nu",
  "malop": "182",
  "ngoaingua": Array (2),
  "monhoc": Array (2)
}
```

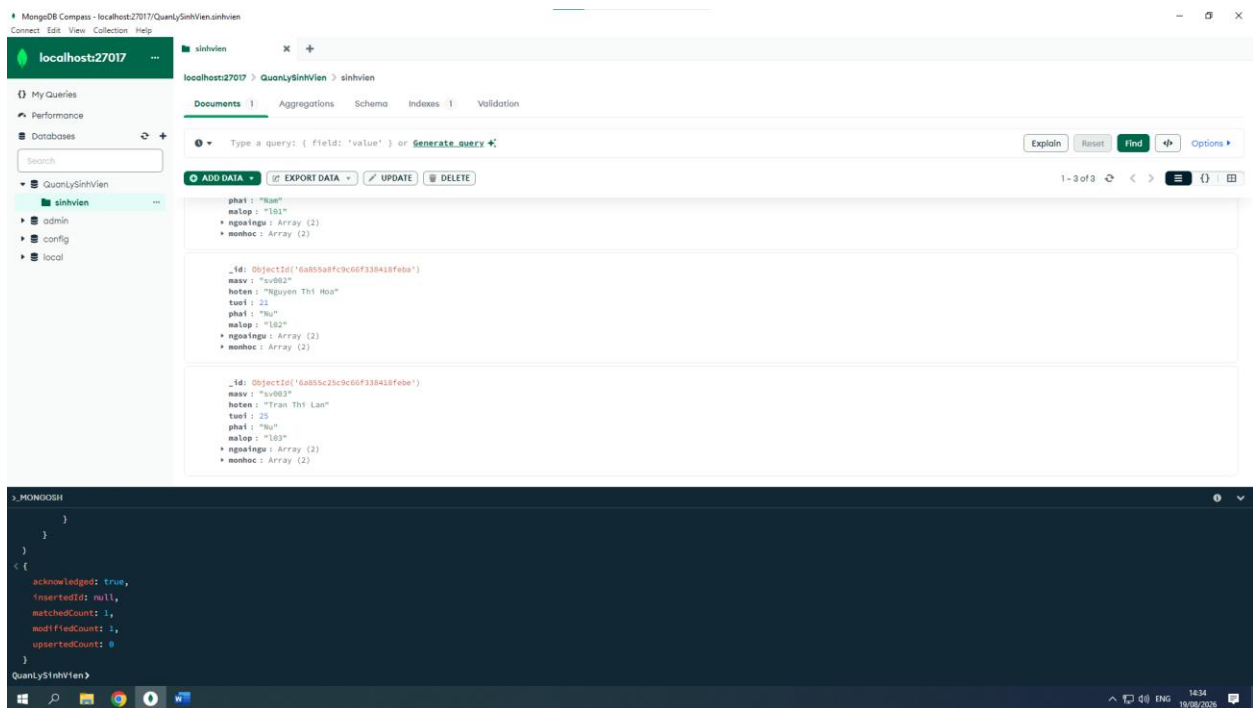
```
> db.sinhvien.deleteMany({
  malop: "183"
})
{
  acknowledged: true,
  deletedCount: 2
}
```

QuanLySinhVien

d)



e)



f)

```

    _MONGODB
  }
}
}
}
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
}
QuantlySinhVlen>

```

g)

```

_MONGODB
    }
}
}
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
}

```

QuantlySinhVien>

Windows taskbar at the bottom shows icons for Start, Search, File Explorer, Chrome, OneDrive, and Word. The system tray on the right displays network status, volume, language (ENG), date (19/08/2026), and time (14:36).

MongoDB Compass - localhost:27017/QuanLySinhVien.sinhvien

localhost:27017

My Queries Performance Databases

QuanLySinhVien

sinhvien

Documents Aggregations Schema Indexes Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA EXPORT DATA UPDATE DELETE

1 - 3 of 3

```
{
  "_id": ObjectId("6a855a86c9c60f338418f6b9"),
  "masv": "sv801",
  "hoten": "Nguyen Van Minh",
  "tuoi": 20,
  "phai": "nam",
  "malop": "101",
  "ngoaingo": Array (2),
  "monhoc": Array (1)
}
```

```
{
  "_id": ObjectId("6a855a86c9c60f338418f6ba"),
  "masv": "sv802",
  "hoten": "Nguyen Thi Hoa",
  "tuoi": 21,
  "phai": "nu",
  "malop": "102",
  "ngoaingo": Array (2),
  "monhoc": Array (2)
}
```

```
{
  "_id": ObjectId("6a855c23c9c60f338418f6bb"),
  "masv": "sv803",
  "hoten": "Tran Thi Lan",
  "tuoi": 20,
  "phai": "nu"
}
```

>_MONGODB

```
[
  ]
}
}
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

QuanLySinhVien>

14:38 19/06/2026